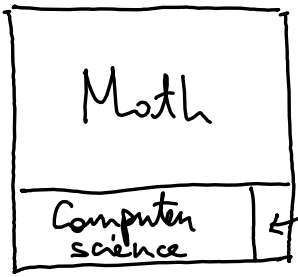
# Lecture 1

Contents of the course:




Prerequisites:
- Linear Algebra
- Probability

## Main directions

1. Experimental
2. Theoretical
   - quantum algorithms
   - quantum information

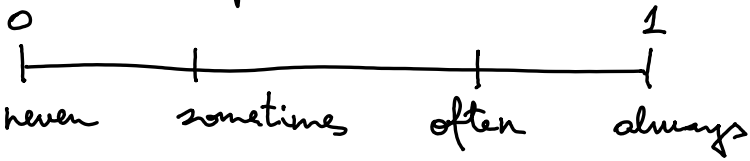
   } mathematical model
3. Practical (quantum programming) ← Cirq

## Motivation

1. The world is quantum
   Information & computation is physical.
   better physics ⇒ better computers
   Quantum mechanics is one of the most powerful and fundamental physical theories.
2. Miniaturization
   Moore's law has slowed down.
   Quantum effects unavoidable.
3. Exploiting quantum phenomena
   Computational advantage (quantum speed-up).
   Note: "classical" = not quantum

# Probabilistic bits (closest thing to quantum bits)

Probability quantifies likelihood:




Probabilistic bit

$$= \begin{cases} 0 & \text{with probability } p_0 \\ 1 & \text{w. p.} \quad p_1 \end{cases}$$

where $p_0, p_1 \geq 0$ and $p_0 + p_1 = 1$.

Example Random coin: $p_0 = p_1 = \frac{1}{2}$.

Note: enough to specify $p_0$ since $p_1 = 1 - p_0$.

Vector notation:

$$\begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = p_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + p_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\textcircled{0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \textcircled{1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Intuition:

$$\textcircled{?} = p_0 \textcircled{0} + p_1 \textcircled{1}$$

$\textcircled{?}$: random coin; $\textcircled{0}$: "heads"; $\textcircled{1}$: "tails"

(deterministic states)

# Probabilistic information is information too!

## Example: Party planning

| Name | Coming? | Chances? |
|---|---|---|
| John | Y | 0.1 |
| Sonah | N | 0.1 |
| Peter | — | 0.8 |
| Anna | — | 0.5 |
| Tom | Y | 0.6 |
| Total: | 2 - 4 | 2.1 |

long range ↑

↑ more informed estimate

# Operations on probabilistic bits

Inverting a coin:

$$\text{NOT}\ (0) = (1) \qquad \text{NOT}\ (1) = (0)$$

Vector notation:

$$\text{NOT}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} \qquad \text{NOT}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$$

How can we extend this to prob. bits?

Extension by linearity:

$$\text{NOT}\left[p_0\begin{pmatrix}1\\0\end{pmatrix} + p_1\begin{pmatrix}0\\1\end{pmatrix}\right] = p_0\,\text{NOT}\begin{pmatrix}1\\0\end{pmatrix} + p_1\,\text{NOT}\begin{pmatrix}0\\1\end{pmatrix}$$

We get:

$$\text{NOT}\begin{pmatrix}p_0\\p_1\end{pmatrix} = p_0\begin{pmatrix}0\\1\end{pmatrix} + p_1\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}p_1\\p_0\end{pmatrix}$$

Intuition: w.p. $p_0$ the bit is (0) and we invert it to (1), and w.p. $p_1$ the bit is (1) and we invert it to (0).

Matrix notation:

$$\text{NOT} = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} \text{ since } \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}\cdot\begin{pmatrix}p_0\\p_1\end{pmatrix} = \begin{pmatrix}p_1\\p_0\end{pmatrix}$$

# Adding and multiplying probabilities

- if you have two independent events and you want to know if they both occurred simultaneously

+ if you have a single event with multiple outcomes and you want to know if any of them occurred (but you don't care which one), so you are grouping them together

Example If I look at my watch at a random point in time, the probability that it is 8:23 is $\frac{1}{12} \cdot \frac{1}{60} = \frac{1}{720}$.

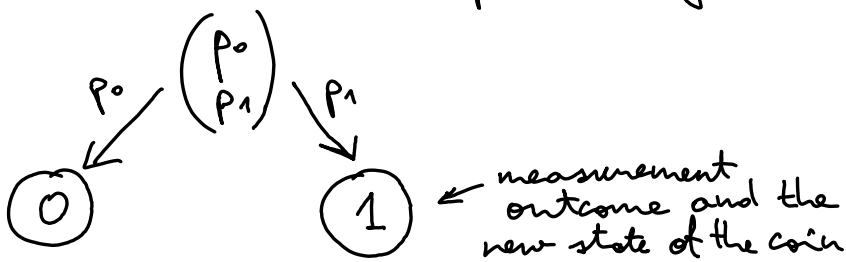
The probability that the hour hand is on a prime is $\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{5}{12}$.

(2 3 5 7 11)

# Measuring a probabilistic bit

Probabilistic bit is like a coin that you have flipped and immediately covered with your hand when it lands. You don't know which side is up. All you know is that each side has the chance of $\frac{1}{2}$ to be up. You can describe such coin by the uniform distribution $\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$.

When you uncover the coin and look at it, this is <u>measurement</u>. You see one of two <u>outcomes</u>, each with a certain probability:




After the measurement, the state of the coin is no longer $\begin{pmatrix} p_0 \\ p_1 \end{pmatrix}$ but either $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ or $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, depending on the measurement outcome. Further measurements will not change the state.

<u>Important</u>: The measurement does not give you the values of $p_0$ and $p_1$. All you get is 0 or 1 and a new state that no longer contains information about $p_0$ and $p_1$.

# Complex numbers

Let $i$ be the imaginary unit ($i^2 = -1$) and $\mathbb{R}$ denote the real numbers, and $\mathbb{C}$ the complex numbers. Any complex number $z \in \mathbb{C}$ can be written in two forms:

- algebraic: $z = a + ib$ $\quad a, b \in \mathbb{R}$
- exponential: $z = r e^{i\varphi}$ $\quad r \geq 0,\ \varphi \in [0, 2\pi)$

Euler's formula is very useful:

$$e^{i\varphi} = \cos\varphi + i \sin\varphi \qquad \varphi \in \mathbb{R}$$

Just like for real numbers,

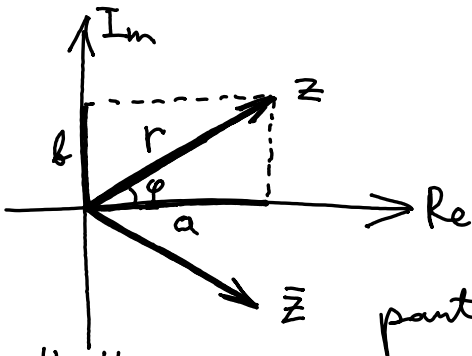
$$e^{i(\alpha+\beta)} = e^{i\alpha} \cdot e^{i\beta} \qquad \alpha, \beta \in \mathbb{R}$$

In particular,

$$e^{i\varphi} \cdot e^{-i\varphi} = e^{i(\varphi-\varphi)} = e^0 = 1$$

so $e^{-i\varphi}$ is the multiplicative inverse of $e^{i\varphi}$.
That means $1/e^{i\varphi} = e^{-i\varphi}$.

# Geometry of $\mathbb{C}$




You can think of $z \in \mathbb{C}$ as a point in the plane. Its coordinates are $(a, b)$ where

$$a = \operatorname{Re} z \qquad b = \operatorname{Im} z$$

are the <u>real</u> and <u>imaginary</u> parts of $z$. Also, $\varphi$ is the angle with the $x$ axis and $r$ is the distance from the origin.

# Arithmetic of $\mathbb{C}$

- addition & subtraction:

$$(a+ib) \pm (c+id) = (a \pm c) + i(b \pm d)$$

- multiplication:

$$(a+ib)\cdot(c+id) = (ac-bd) + i(ad+bc)$$

$$r_1 e^{i\varphi_1} \cdot r_2 e^{i\varphi_2} = r_1 r_2 \, e^{i(\varphi_1+\varphi_2)}$$

- absolute value:

$$|z| = \sqrt{a^2+b^2} = r \qquad |z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

Squared absolute value plays an important role:

$$|z|^2 = a^2 + b^2 = r^2$$

- complex conjugation: $\overline{i} = -i$

$$\bar{z} = z^* = a - ib = r e^{-i\varphi}$$

very useful for computing squared absolute value:

$$z \cdot \bar{z} = |z|^2$$

and the real and imaginary parts:

$$z + \bar{z} = 2 \operatorname{Re} z \qquad z - \bar{z} = 2i \operatorname{Im} z$$

# Quantum bits (qubits)

Only two differences between probabilistic and quantum bits:

1. probabilities replaced by complex amplitudes,
2. amplitudes are squared when measuring.

Everything else in quantum computing is a consequence of these two small changes!

When added, complex (or just negative) numbers can cancel — this important effect is called interference and it does not happen with probabilities.

## Qubit states

Instead of coins (0) and (1) we use "kets":

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A general qubit state is a superposition (or linear combination) of these:

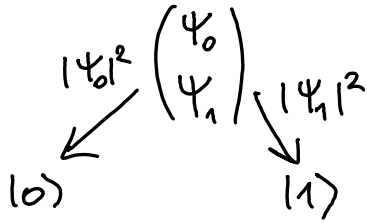
$$\psi_0|0\rangle + \psi_1|1\rangle = \psi_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \psi_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}$$

where $\psi_0, \psi_1 \in \mathbb{C}$ are called amplitudes and satisfy

$$|\psi_0|^2 + |\psi_1|^2 = 1$$

# Measuring a qubit

**Born rule:** probability = |amplitude|$^2$




If the measurement outcome is $k \in \{0,1\}$ then the state after the measurement becomes $|k\rangle$ and the original state is gone. In particular, you don't learn $\psi_0$ or $\psi_1$ this way.

# Quantum NOT

Just like for probabilistic bits, quantum operations are <u>extended by linearity</u>:

$$\begin{aligned}\text{NOT}\left(\psi_0|0\rangle + \psi_1|1\rangle\right) &= \psi_0\,\text{NOT}|0\rangle + \psi_1\,\text{NOT}|1\rangle \\ &= \psi_0|1\rangle + \psi_1|0\rangle.\end{aligned}$$

The output is also a quantum state since $|\psi_0|^2 + |\psi_1|^2 = 1$.

# General qubit operation

Let's call it $U$. For some $a,b,c,d \in \mathbb{C}$,

$$U|0\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{and} \quad U|1\rangle = \begin{pmatrix} c \\ d \end{pmatrix}$$

These too are quantum states so

$$|a|^2 + |b|^2 = 1 \quad \text{and} \quad |c|^2 + |d|^2 = 1$$

Are these the only constraints? Let $|\psi\rangle := \psi_0|0\rangle + \psi_1|1\rangle$ be an arbitrary qubit state ($|\psi_0|^2 + |\psi_1|^2 = 1$). By linearity,

$$U|\psi\rangle = \psi_0 U|0\rangle + \psi_1 U|1\rangle = \psi_0 \begin{pmatrix} a \\ b \end{pmatrix} + \psi_1 \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} \psi_0 a + \psi_1 c \\ \psi_0 b + \psi_1 d \end{pmatrix} = \begin{pmatrix} \psi_0' \\ \psi_1' \end{pmatrix}$$

This should also be a quantum state so

$$|\psi_0'|^2 + |\psi_1'|^2 = |\psi_0 a + \psi_1 c|^2 + |\psi_0 b + \psi_1 d|^2$$

$$= (\psi_0 a + \psi_1 c)(\bar{\psi}_0 \bar{a} + \bar{\psi}_1 \bar{c}) + (\psi_0 b + \psi_1 d)(\bar{\psi}_0 \bar{b} + \bar{\psi}_1 \bar{d})$$

$$= |\psi_0|^2 \underbrace{(|a|^2 + |b|^2)}_{1} + |\psi_1|^2 \underbrace{(|c|^2 + |d|^2)}_{1}$$

$$+ \psi_0 \bar{\psi}_1 \underbrace{(a\bar{c} + b\bar{d})}_{0} + \psi_1 \bar{\psi}_0 \underbrace{(c\bar{a} + d\bar{b})}_{0}$$

$$= |\psi_0|^2 + |\psi_1|^2 \quad \text{for all } \psi_0, \psi_1 \in \mathbb{C}.$$

Recall: $z \cdot \bar{z} = |z|^2$

$a\bar{c} + b\bar{d}$ so it's redundant

A general qubit operation is of the form $U = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ where $a,b,c,d \in \mathbb{C}$ are such that

$$|a|^2 + |b|^2 = 1, \quad |c|^2 + |d|^2 = 1, \quad a\bar{c} + b\bar{d} = 0.$$

# Global and relative phases

- The phase of $re^{i\varphi}$ is $e^{i\varphi}$. We say that two vectors $|\psi\rangle$ and $|\psi'\rangle$ differ only by a global phase if

$$|\psi\rangle = e^{i\varphi}|\psi'\rangle \qquad \text{for some } \varphi \in \mathbb{R}.$$

Such states produce the same measurement statistics since $|e^{i\varphi}z| = |z|$ for any $z \in \mathbb{C}$. Moreover, $U|\psi\rangle = e^{i\varphi}U|\psi'\rangle$ so they still differ only by a global phase also after any operation $U$. This means that $|\psi\rangle$ and $|\psi'\rangle$ represent the same physical reality.

- For comparison, consider

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

They also produce the same measurement statistics: $\begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$

Are $|+\rangle$ and $|-\rangle$ the same state? No.
In fact, you can find $U$ such that $U|+\rangle = |0\rangle$ and $U|-\rangle = |1\rangle$
Generally, the $e^{i\varphi}$ in $|0\rangle + e^{i\varphi}|1\rangle$ is called a relative phase.

Remember: global phase doesn't matter, relative phase matters!

# Qubit geometry

Since we can ignore the global phase, any qubit state can be written as

$$|\psi\rangle = \underbrace{\cos\frac{\theta}{2}}_{\psi_0}|0\rangle + \underbrace{e^{i\varphi}\sin\frac{\theta}{2}}_{\psi_1}|1\rangle$$

for some angles $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$.

Note that $\psi_0 \in [0,1]$ and $|\psi_0|^2 + |\psi_1|^2 = 1$.

Also: $\theta = 0 \Rightarrow |\psi\rangle = |0\rangle$

$\theta = \pi \Rightarrow |\psi\rangle = e^{i\varphi}|1\rangle \sim |1\rangle$ ("up to global phase")

In these two extreme cases the parameter $\varphi$ has become redundant. This is similar to how the longitude become irrelevant at the poles of the Earth, suggesting that qubit has the same topology as the sphere! Indeed, there is a one-to-one correspondence between qubit states $|\psi\rangle \in \mathbb{C}^2$ and unit vectors in $\mathbb{R}^3$:

$$\begin{pmatrix}\psi_0\\ \psi_1\end{pmatrix} \in \mathbb{C}^2 \longrightarrow \begin{pmatrix}2\,\mathrm{Re}(\bar{\psi}_0\psi_1)\\ 2\,\mathrm{Im}(\bar{\psi}_0\psi_1)\\ |\psi_0|^2 - |\psi_1|^2\end{pmatrix} \in \mathbb{R}^3$$

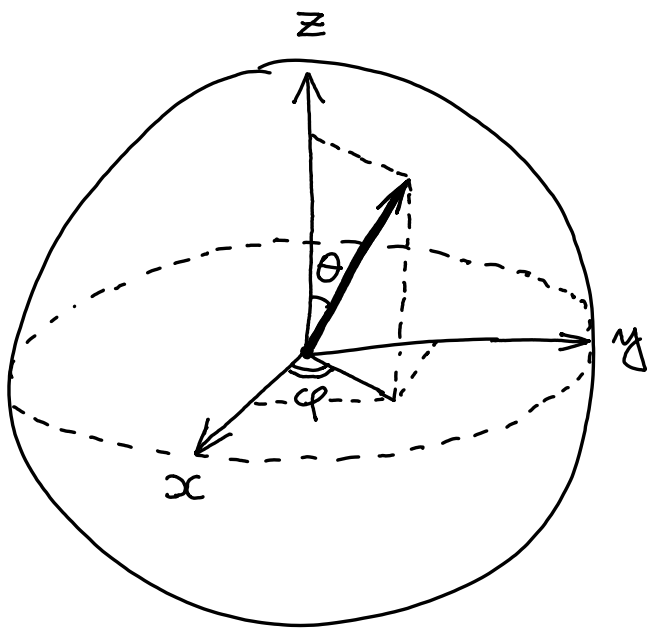
The resulting vector in $\mathbb{R}^3$ is called the Bloch vector of $|\psi\rangle$.

Going the other way:

$$\frac{1}{\sqrt{2(z+1)}}\begin{pmatrix} z+1 \\ x+iy \end{pmatrix} \in \mathbb{C}^2 \longleftarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$$

## Bloch sphere

The correspondence is particularly elegant when expressed using the spherical coordinates:




$$\begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi}\sin\frac{\theta}{2} \end{pmatrix} \longrightarrow \begin{cases} x = \sin\theta\cos\varphi \\ y = \sin\theta\sin\varphi \\ z = \cos\theta \end{cases}$$